Flow charts and operations

1

$$s \xrightarrow{-7} s-7 \xrightarrow{\times 6} 6(s-7)$$

2

$$t \xrightarrow{+3} t+3 \xrightarrow{\div 5} \frac{(t+3)}{5}$$

3

a $$y \xrightarrow{\times 5} 5y \xrightarrow{+8} 5y+8$$

b $$k \xrightarrow{\div 3} \frac{k}{3} \xrightarrow{-7} \frac{k}{3}-7$$

c $$p \xrightarrow{+15} p+15 \xrightarrow{\times 8} 8(p+15)$$

d $$n \xrightarrow{+5} n+5 \xrightarrow{\div 3} \frac{n+5}{3}$$

e $$x \xrightarrow{\times 4} 4x \xrightarrow{+3} 4x+3$$

f $$j \xrightarrow{\div 9} \frac{j}{9} \xrightarrow{-7} \frac{j}{9}-7$$

g $$p \xrightarrow{+28} p+28 \xrightarrow{\times 3} 3(p+28)$$

h $$s \xrightarrow{-6} s+5 \xrightarrow{\times 7} 7(s-6)$$

Operations in reverse order

4

a $$6y+9 \xrightarrow{-9} 6y \xrightarrow{\div 6} y$$

b $$5(q+3) \xrightarrow{\div 5} q+3 \xrightarrow{-3} q$$

c $$6(s-9) \xrightarrow{\div 6} s-9 \xrightarrow{+9} s$$

d $$\frac{n+8}{5} \xrightarrow{\times 5} n+8 \xrightarrow{-8} n$$

e $$\frac{j}{5}-6 \xrightarrow{+6} \frac{j}{5} \xrightarrow{\times 5} j$$

f $$8y+9 \xrightarrow{-9} 8y \xrightarrow{\div 8} y$$

5

a $4(q+9) \xrightarrow{\div 4} q+9 \xrightarrow{-9} q$

b $5(r-7) \xrightarrow{\div 5} r-7 \xrightarrow{+7} r$

c $\dfrac{n+6}{9} \xrightarrow{\times 9} n+6 \xrightarrow{-6} n$

d $\dfrac{j}{2}-7 \xrightarrow{+7} \dfrac{j}{2} \xrightarrow{\times 2} j$

Two-step equations

6

a Operation 1: Subtract 16

Operation 2: Divide by 4

b $24 \xrightarrow{\text{Operation 1}} 8 \xrightarrow{\text{Operation 2}} 2$

c $x = 2$

7

a Operation 1: Divide by 3

Operation 2: Subtract 11

b $57 \xrightarrow{\text{Operation 1}} 19 \xrightarrow{\text{Operation 2}} 8$

c $p = 8$

8

a Operation 1: Add 11

Operation 2: Multiply by 7

b $-7 \xrightarrow{\text{Operation 1}} 4 \xrightarrow{\text{Operation 2}} 28$

c $j = 28$

9

a

i Subtract 6 from both sides, then divide both sides by 7.

ii $p = 5$

b

i Add 5 to both sides, then divide both sides by 2.

ii $x = 8$

c

i Divide both sides by 5, then add 15 to both sides.

ii $n = 22$

d

i Divide both sides by 10, then subtract 10 from both sides.

ii $p = 2$

e

i Subtract 3 from both sides, then multiply both sides by 8.

ii $y = 16$

f

i Add 11 to both sides, then multiply both sides by 3.

ii $x = 60$

g

i Multiply both sides by 6, then subtract 22 from both sides.

ii $u = 2$

h

i Multiply both sides by 2, then add 10 to both sides.

ii $y = 20$